

3212. Count Submatrices With Equal Frequency of X and Y

Difficulty: Medium

Link: <https://leetcode.com/problems/count-submatrices-with-equal-frequency-of-x-and-y/description/?envType=daily-question&envId=2026-03-19>

Generated at: 2026-03-19 10:06:40

Problem:

Given a 2D character matrix `grid`, where `grid[i][j]` is either `'X'`, `'Y'`, or `'.'`, return the number of submatrices that contain:

- `grid[0][0]`
- an **equal** frequency of `'X'` and `'Y'`.
- **at least** one `'X'`.

Example 1:

Input: `grid = [["X","Y","."],["Y",".","."]]`

Output: 3

Explanation:

X	Y	•
Y	•	•

X	Y	•
Y	•	•

X	Y	•
Y	•	•

Example 2:

Input: grid = `[["X","X"],["X","Y"]]`

Output: 0

Explanation:

No submatrix has an equal frequency of 'X' and 'Y'.

Example 3:

Input: `grid = [[".", ".", "."], [".", ".", "."]]`

Output: 0

Explanation:

No submatrix has at least one 'X' .

Constraints:

- `1 <= grid.length, grid[i].length <= 1000`
- `grid[i][j]` is either 'X' , 'Y' , or '.' .